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# **BUSINESS REQUIREMENTS**

**for**

## **LKFORUM WEBSITE**

**Version 1.0**

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**May 24, 2026**

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# 1 Objective and Scope

## 1.1 Project Objective

The primary objective of the LKForum project is to design and build a modern, scalable, and highly interactive community platform. Unlike general social networks, LKForum focuses on autonomous community management, allowing users to create, configure, and moderate their own niche sub-forums (Communities).

The development focuses on delivering a high-quality system that ensures:

- High performance for real-time interactions, including Instant Messaging and Notifications.
- Superior scalability to handle a large and growing number of concurrent communities and users.
- Robust moderation capabilities, integrating both automated (AI-assisted) and manual review processes.
- High standards for security and data privacy to protect user information and platform integrity.

## 1.2 Project Scope

The project scope includes the following modules:

- **Identity & Access Management:** Multi-factor authentication, SSO, and granular role-based access control.
- **Community Management:** Creation, configuration (public/private/18+), rule-setting, and membership handling.
- **Content Management:** Multi-format posts (text, image, video, poll), nested comments, and draft management.
- **Engagement System:** Real-time voting (upvote/downvote), saved posts, and activity history.
- **Communication Suite:** Real-time 1-1 messaging via WebSockets and an instant notification system.
- **Moderation & Governance:** Report handling, user banning/muting, and content approval workflows.
- **Administration Dashboard:** Global system oversight, analytics, and platform-wide moderation.

## 2 Business Requirements

### 2.1 Application Overview

LKForum is a community-driven social network structured around “Communities” (sub-forums).

- **Users** can discover and join communities based on interests.
- **Content** is categorized by communities, ensuring relevant discussions.
- **Decentralized Moderation:** Each community has its own set of rules and a dedicated moderation team (Owners/Moderators), reducing the burden on global administrators.
- **Real-time Experience:** The platform provides instant feedback, live messaging, and notifications.

### 2.2 Domain Model

#### 2.2.1 Domain Diagram (Conceptual)

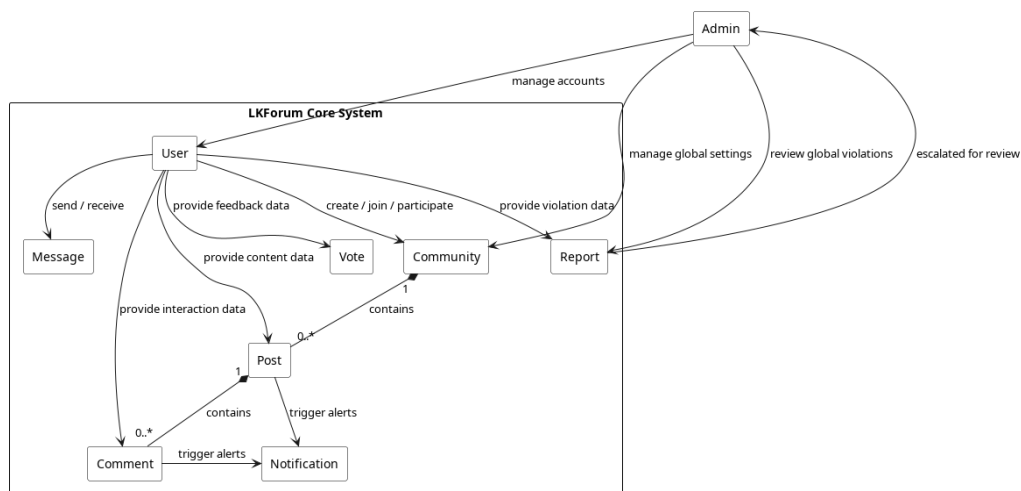


Figure 2.1: Domain Model Diagram

## 2.2.2 Domain Objects Description

| # | Object Name  | Object Description                                                        |
|---|--------------|---------------------------------------------------------------------------|
| 1 | User         | A registered identity with a profile, karma points, and activity history. |
| 2 | Community    | A niche hub with specific topics, rules, and privacy settings.            |
| 3 | Post         | Content shared within a community (Text, Image, Video, or Poll).          |
| 4 | Comment      | A response to a post, supporting nested/threaded replies.                 |
| 5 | Vote         | An expression of approval (Upvote) or disapproval (Downvote) on content.  |
| 6 | Message      | Direct communication content between users in a 1-1 chat.                 |
| 7 | Notification | Real-time alerts for replies, mentions, votes, or system messages.        |
| 8 | Report       | A user-generated complaint about content or behavior violating rules.     |

Table 2.1: Domain Objects Description

## 2.3 Workflow

### 2.3.1 Content Lifecycle & Moderation Workflow

1. **Creation:** A user creates a post in a community.
2. **AI Screening:** The system automatically checks content for prohibited keywords/media.
3. **Approval (Optional):** If the community requires post approval, the post is held in a *Pending* state.
4. **Moderation:** A Moderator/Owner reviews the post (Approve or Reject).
5. **Publication:** Once approved, the post becomes public and triggers notifications to members.

### 2.3.2 Community Management Workflow

1. **Setup:** A user creates a community and becomes the Owner.
2. **Configuration:** The Owner sets rules, privacy settings, and invites Moderators.
3. **Governance:** Owners and Moderators monitor reports and can ban/mute users or remove content.

2.4 Use Cases and Actors

2.4.1 Use Case Diagram

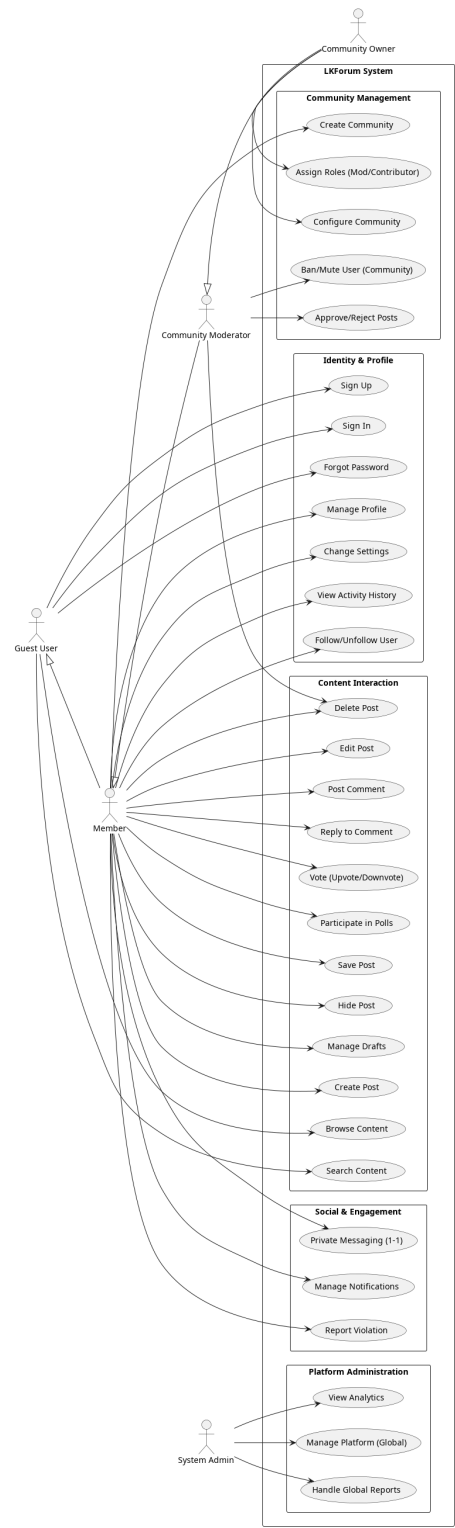


Figure 2.2: Use Case Diagram

### 2.4.2 Description of Actors

| ID  | Actor Name          | Definition                                                            |
|-----|---------------------|-----------------------------------------------------------------------|
| U-1 | Guest User          | Unauthenticated user; can browse public content and search.           |
| U-2 | Member              | Authenticated user; can post, comment, vote, and message.             |
| U-3 | Community Moderator | Member with management rights in a specific community.                |
| U-4 | Community Owner     | Creator of a community; can appoint moderators and change settings.   |
| U-5 | System Admin        | Global administrator responsible for platform-wide health and safety. |

Table 2.2: Description of Actors

### 2.4.3 Description of Use Cases

| ID    | Use Case Name                | Definition                                                                               |
|-------|------------------------------|------------------------------------------------------------------------------------------|
| UC-01 | <b>Sign Up</b>               | Users create a new account by providing email, username, and password, verified via OTP. |
| UC-02 | <b>Sign In</b>               | Authenticated access using credentials or linked Google OAuth account.                   |
| UC-03 | <b>Forgot Password</b>       | Users reset their password through email verification (OTP).                             |
| UC-04 | <b>Manage Profile</b>        | Users update personal information, including bio, avatar, and cover image.               |
| UC-05 | <b>Change Settings</b>       | Users configure account preferences (privacy, notifications, theme).                     |
| UC-06 | <b>View Activity History</b> | Users review their past posts, comments, and interactions.                               |
| UC-07 | <b>Follow/Unfollow User</b>  | Users subscribe to another user's activity feed.                                         |
| UC-08 | <b>Browse Content</b>        | Users explore communities and view public posts without necessarily logging in.          |
| UC-09 | <b>Search Content</b>        | Users search for specific keywords, posts, users, or communities.                        |
| UC-10 | <b>Create Post</b>           | Members publish new content (Text, Image, Video) within a selected community.            |

Table 2.3: Description of Use Cases

|       |                              |                                                                              |
|-------|------------------------------|------------------------------------------------------------------------------|
| UC-11 | <b>Edit Post</b>             | Members modify the content of their existing posts (stores edit history).    |
| UC-12 | <b>Delete Post</b>           | Members or Moderators remove posts from public view (soft delete).           |
| UC-13 | <b>Post Comment</b>          | Users add a response directly to a post.                                     |
| UC-14 | <b>Reply to Comment</b>      | Users respond to an existing comment (supporting nested threads).            |
| UC-15 | <b>Vote on Content</b>       | Users express approval (Upvote) or disapproval (Downvote) on posts/comments. |
| UC-16 | <b>Participate in Poll</b>   | Users cast votes in community-created polls.                                 |
| UC-17 | <b>Save Post</b>             | Users bookmark posts for quick access later.                                 |
| UC-18 | <b>Hide Post</b>             | Users hide specific posts from their personal feed.                          |
| UC-19 | <b>Manage Drafts</b>         | Users save unfinished posts as drafts and resume editing later.              |
| UC-20 | <b>Private Messaging</b>     | Users send and receive instant 1-1 messages via WebSocket.                   |
| UC-21 | <b>Manage Notifications</b>  | Users view and clear real-time alerts for engagement and system updates.     |
| UC-22 | <b>Report Violation</b>      | Users flag inappropriate content or behavior for moderator review.           |
| UC-23 | <b>Create Community</b>      | Users establish a new niche forum and become the “Owner.”                    |
| UC-24 | <b>Approve/Reject Post</b>   | Moderators review pending posts before they appear in the community.         |
| UC-25 | <b>Ban/Mute User</b>         | Moderators restrict a user’s ability to participate within a community.      |
| UC-26 | <b>Configure Community</b>   | Owners adjust rules, privacy settings, and appearance of their community.    |
| UC-27 | <b>Assign Roles</b>          | Owners appoint other members as Moderators or Contributors.                  |
| UC-28 | <b>Handle Global Reports</b> | System Admins review and act on platform-wide violation reports.             |
| UC-29 | <b>Manage Platform</b>       | System Admins ban/unban users or communities globally.                       |
| UC-30 | <b>View Analytics</b>        | System Admins monitor platform statistics (DAU, growth, content volume).     |

Table 2.3: Description of Use Cases



## 2.5 Security Matrix

| #  | Function                            | Guest | Member | Mod | Owner | Admin |
|----|-------------------------------------|-------|--------|-----|-------|-------|
| 1  | Sign Up / Sign In / Forgot Password | X     | X      | X   | X     | X     |
| 2  | Manage Profile & Settings           |       | X      | X   | X     | X     |
| 3  | Follow / Unfollow User              |       | X      | X   | X     | X     |
| 4  | Browse Public Content & Search      | X     | X      | X   | X     | X     |
| 5  | View Own Activity History           |       | X      | X   | X     | X     |
| 6  | Create Post / Comment / Reply       |       | X      | X   | X     | X     |
| 7  | Edit / Delete Own Content           |       | X      | X   | X     | X     |
| 8  | Vote / Participate in Polls         |       | X      | X   | X     | X     |
| 9  | Save / Hide Post / Manage Drafts    |       | X      | X   | X     | X     |
| 10 | Send/Receive Private Messages       |       | X      | X   | X     | X     |
| 11 | Receive Notifications               |       | X      | X   | X     | X     |
| 12 | Report Violations                   |       | X      | X   | X     | X     |
| 13 | Create New Community                |       | X      | X   | X     | X     |
| 14 | Approve / Reject / Pin Posts        |       |        | X   | X     | X     |
| 15 | Delete Any Post (In Community)      |       |        | X   | X     | X     |
| 16 | Ban / Mute User (In Community)      |       |        | X   | X     | X     |
| 17 | Configure Community Settings        |       |        |     | X     | X     |
| 18 | Assign Roles (Mod/Contributor)      |       |        |     | X     | X     |
| 19 | Handle Global Reports               |       |        |     |       | X     |
| 20 | Global User / Community Ban         |       |        |     |       | X     |
| 21 | View System Analytics               |       |        |     |       | X     |

Table 2.4: Security Matrix

## 2.6 User Story

### 2.6.1 Guest Users

| ID    | User Story                                                                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| US-01 | As a Guest User, I want to browse posts so that I can explore content without creating an account.                                             |
| US-02 | As a Guest User, I want to view posts and comments so that I can read discussions.                                                             |
| US-03 | As a Guest User, I want to search for keywords so that I can find relevant content.                                                            |
| US-04 | As a Guest User, I want to search for communities so that I can discover topics of interest.                                                   |
| US-05 | As a Guest User, I want to view user profiles so that I can learn about content creators.                                                      |
| US-06 | As a Guest User, I want to be prompted to sign up when attempting restricted actions so that I understand the benefits of creating an account. |
| US-07 | As a Guest User, I want to sign up with my email and verify via OTP so that I can securely create an account.                                  |

Table 2.5: Guest User Stories

### 2.6.2 Members

| ID    | User Story                                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| US-08 | As a Member, I want to sign in using my Google account so that I can quickly access the platform without creating a new password. |
| US-09 | As a Member, I want to sign in using my email and password so that I can access my account securely.                              |
| US-10 | As a Member, I want to customize my profile with a bio so that I can describe myself.                                             |
| US-11 | As a Member, I want to upload an avatar so that other users can recognize me.                                                     |
| US-12 | As a Member, I want to upload a profile banner so that I can personalize my profile page.                                         |
| US-13 | As a Member, I want to create a text post so that I can share content with others.                                                |
| US-14 | As a Member, I want to upload images to my post so that I can share visual content.                                               |
| US-15 | As a Member, I want to create a poll in my post so that I can gather opinions from others.                                        |

Table 2.6: Member User Stories

|       |                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------|
| US-16 | As a Member, I want to save my post as a draft so that I can edit and publish it later.                         |
| US-17 | As a Member, I want to upvote posts so that valuable content is highlighted.                                    |
| US-18 | As a Member, I want to downvote posts so that low-quality content is less visible.                              |
| US-19 | As a Member, I want to upvote comments so that helpful responses are recognized.                                |
| US-20 | As a Member, I want to downvote comments so that unhelpful responses are less visible.                          |
| US-21 | As a Member, I want to create, modify, and delete comments under a post.                                        |
| US-22 | As a Member, I want to reply to a specific comment so that conversations remain structured.                     |
| US-23 | As a Member, I want to receive notifications when someone replies to my post or comment so that I stay updated. |
| US-24 | As a Member, I want to receive notifications when someone mentions me so that I can respond quickly.            |
| US-25 | As a Member, I want to follow other users so that I can see their activities in my feed.                        |
| US-26 | As a Member, I want to view posts from users I follow so that I can stay updated with their content.            |
| US-27 | As a Member, I want to send private messages to other users so that I can communicate one-on-one.               |
| US-28 | As a Member, I want to report inappropriate content so that I can help maintain a safe community.               |

Table 2.6: Member User Stories

### 2.6.3 Community Owners & Moderators

| ID    | User Story                                                                                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------|
| US-29 | As a Community Owner, I want to create a community so that I can build a space for specific topics.                     |
| US-30 | As a Community Owner, I want to define community rules so that members understand acceptable behavior.                  |
| US-31 | As a Community Owner, I want to set privacy settings (public or private) so that I can control access to the community. |
| US-32 | As a Community Owner, I want to assign moderator roles to members so that they can help manage the community.           |
| US-33 | As a Community Moderator, I want to review and approve pending posts so that only appropriate content is published.     |

Table 2.7: Community Owner and Moderator User Stories

|       |                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------|
| US-34 | As a Community Moderator, I want to remove posts or comments so that I can enforce community rules. |
| US-35 | As a Community Moderator, I want to mute users so that they temporarily cannot interact.            |
| US-36 | As a Community Moderator, I want to ban users so that I can prevent disruptive behavior.            |

Table 2.7: Community Owner and Moderator User Stories

#### 2.6.4 System Administrators

| ID    | User Story                                                                                                      |
|-------|-----------------------------------------------------------------------------------------------------------------|
| US-37 | As a System Admin, I want to view a dashboard with system analytics so that I can monitor platform performance. |
| US-38 | As a System Admin, I want to review reports escalated from communities so that I can take appropriate action.   |
| US-39 | As a System Admin, I want to ban users globally so that malicious users are removed from the platform.          |
| US-40 | As a System Admin, I want to remove or restrict communities so that harmful communities are controlled.         |

Table 2.8: System Administrator User Stories

## 3 Appendix

### 3.1 Glossary

- **Karma:** A reputation score based on the net value of upvotes and downvotes received.
- **Sub-forum:** Synonymous with “Community” in LKForum.
- **SSO:** Single Sign-On (Google Authentication).
- **OTP:** One-Time Password sent via email for verification.

### 3.2 Open Issues

- Implementation of cross-posting (sharing posts across multiple communities).
- Automated AI moderation for image/video content (currently focusing on text).
- Integration of a global search engine (Elasticsearch).